



ANALYSTLAB AFRICA

Data Science Internship Programme

Official Weekly Assignment

Week 3: Advanced Data Analysis, Statistical Validation & Feature Engineering

Document Control

Programme	Data Science Internship
Week	Week 1
Project	Advanced Data Analysis, Statistical Validation & Feature Engineering
Version	1.0
Issue Date	_____
Submission Deadline	Sunday, 11:59 PM (WAT)

Business Scenario

You are continuing your role as a **Junior Data Scientist at AnalystLab Africa Consulting**.

During **Week 1**, you defined your business problem, explored your dataset, identified initial patterns, and assessed data-quality issues.

During **Week 2**, you cleaned, prepared, and structured your dataset for advanced analysis and future machine learning.

In Week 3, your responsibility is to conduct a **deeper investigation of the dataset** before predictive modelling begins.

You will use the cleaned and prepared dataset from Week 2 to:

- Investigate more complex relationships.
- Validate important assumptions using statistical analysis.
- Perform hypothesis testing.
- Engineer meaningful features where appropriate.
- Evaluate the usefulness of existing and engineered features.
- Refine your dataset based on analytical evidence.
- Translate your findings into meaningful business recommendations.

This is a continuation of your existing project.

You should continue using:

- The same business problem.
- The same dataset.
- The same project repository.
- The cleaned and prepared dataset from Week 2.

Do not restart the project or repeat the complete data cleaning process from Week 2.

You should only revisit a cleaning or preprocessing decision if your Week 3 analysis provides a valid reason to do so.

The work completed this week will prepare your project for:

Week 4: Machine Learning Model Development, Evaluation & Business Recommendations

2. LEARNING OBJECTIVES

By the end of this assignment, you should be able to:

- Perform advanced exploratory data analysis.
- Investigate complex relationships between multiple variables.
- Formulate research and business hypotheses.
- Select appropriate statistical tests.
- Interpret test statistics and p-values.
- Translate statistical findings into business implications.
- Engineer meaningful features using analytical and domain knowledge.
- Evaluate the usefulness of original and engineered features.
- Identify redundant or potentially unhelpful features.
- Refine a dataset for future machine learning.
- Communicate findings professionally to technical and non-technical stakeholders.

3. TOOLS TO USE

Required Tools

You should use the following tools where applicable:

- Python
- Jupyter Notebook or Google Colab
- GitHub
- Google Drive

Core Python Libraries

You are expected to use the following libraries where appropriate:

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Statistical Analysis Libraries

You may use:

- SciPy
- Statsmodels

Optional Libraries and Tools

You may also use:

- Plotly
- Missingno
- Imbalanced-learn
- Visual Studio Code
- JupyterLab

You are not required to use every tool or library listed above. Select tools and techniques that are appropriate for your dataset and clearly justify your choices.

4. ASSIGNMENT TASKS

PART 1: REVIEW OF PREVIOUS WORK

Begin by reviewing the work completed during Weeks 1 and 2.

Prepare a brief summary covering:

- The business problem.
- The main project objective.
- The potential or confirmed target variable.
- Key findings from Week 1.
- Major data-quality issues identified in Week 1.
- Major cleaning and preprocessing actions completed in Week 2.
- Remaining questions that require deeper investigation.

Deliverable

Prepare a **Project Continuity Summary**.

This should briefly demonstrate how your Week 3 analysis builds on your previous work.

PART 2: ADVANCED EXPLORATORY DATA ANALYSIS

Perform a deeper level of exploratory data analysis.

The purpose of this stage is to generate **new insights**, not simply repeat the complete Week 1 EDA.

Your analysis may include the following:

Numerical Analysis

Investigate:

- Distributions.
- Skewness.
- Variability.
- Potential unusual patterns.
- Relationships with other important variables.

You may use:

- Histograms.
- Density plots.
- Box plots.
- Violin plots.
- Scatter plots.

Categorical Analysis

Investigate:

- Frequency distributions.
- Percentages.
- Group comparisons.
- Relationships with the target variable, where applicable.

You may use:

- Bar charts.
- Count plots.
- Stacked bar charts.
- Cross-tabulations.

Bivariate Analysis

Investigate important relationships between:

- Numerical and numerical variables.
- Numerical and categorical variables.
- Categorical and categorical variables.

Focus on relationships that may help answer your business questions.

Multivariate Analysis

Investigate relationships involving multiple variables.

You may use:

- Correlation analysis.
- Pair plots or equivalent visualisations.
- Grouped comparisons.
- Faceted charts.
- Multivariable scatter plots.
- Pivot tables or grouped summaries.

Deliverable

Document your **Advanced EDA Findings** in your notebook.

Focus on insights that were not fully explored during Week 1.

PART 3: STATISTICAL HYPOTHESIS TESTING AND VALIDATION

Perform at least **four appropriate statistical analyses or tests** based on your dataset and business questions.

Possible methods include:

- Correlation analysis.
- Chi-Square Test.
- Independent Samples T-Test.
- Paired T-Test.
- ANOVA.
- Mann-Whitney U Test.
- Kruskal-Wallis Test.
- Regression analysis.
- Other appropriate statistical techniques.

You are **not required to use every test listed above**.

Your selected tests must be appropriate for:

- The type of variables being analysed.
- The distribution of the data.
- Your research or business question.

For each statistical analysis, include:

1. The business or research question.
2. The objective of the test.
3. Null Hypothesis (H_0).
4. Alternative Hypothesis (H_1).
5. The statistical method selected.
6. Justification for selecting the method.
7. Test statistic, where applicable.
8. P-value, where applicable.
9. Decision regarding the hypothesis.
10. Clear interpretation of the result.
11. Business implication of the finding.

Deliverable

Prepare a **Statistical Analysis Summary**.

Your interpretations should focus on what the results mean, rather than simply reporting statistical values.

PART 4: ADVANCED FEATURE ENGINEERING

Create meaningful features that may improve future analysis or predictive modelling.

Feature engineering should be based on:

- Domain knowledge.
- Findings from your exploratory analysis.
- Statistical evidence.
- Business understanding.

Examples may include:

Numerical Features

- Ratios.
- Percentages.
- Aggregated values.
- Interaction variables.
- Log transformations.
- Binned numerical variables.

Categorical Features

- Age groups.
- Income bands.
- Risk categories.
- Customer segments.
- Employment categories.

The features you create must be meaningful and appropriate for your selected dataset.

For each engineered feature, explain:

- How the feature was created.
- Why it was created.
- Its business meaning.
- Its potential value for future analysis or modelling.

Deliverable

Create at least **three meaningful engineered features**, where appropriate for your dataset.

Document all feature engineering decisions.

PART 5: FEATURE EVALUATION AND SELECTION

Evaluate both your original and newly engineered features.

Your objective is to determine which variables may provide useful information for future modelling.

Investigate:

- Correlation between numerical variables.
- Highly correlated variables.
- Potential multicollinearity.
- Redundant variables.
- Features with limited analytical value.
- Relationships between features and the target variable.
- Potential data leakage.

You may use:

- Correlation analysis.
- Statistical evidence.
- Mutual information.
- Feature importance from an appropriate preliminary model.
- Variance analysis.
- Domain knowledge.

You are not expected to remove a feature simply because it is correlated with another feature. Every decision should be justified.

For each important decision, explain:

- Which feature was evaluated.
- The method used.
- The evidence obtained.
- Whether the feature was retained or removed.
- Why the decision was appropriate.

Deliverable

Prepare a **Feature Evaluation and Selection Summary**.

PART 6: DATASET REFINEMENT

Use the findings from your advanced analysis and feature evaluation to refine your Week 2 dataset.

You may:

- Add meaningful engineered features.
- Remove justified redundant features.
- Correct newly discovered issues.
- Update transformations where analytical evidence supports the change.

You should **not repeat the complete Week 2 cleaning and preprocessing process**.

All significant changes must be documented and justified.

Deliverable

Produce a **Final Modelling Dataset**.

This dataset will serve as the foundation for Week 4 Machine Learning.

PART 7: BUSINESS INSIGHTS AND RECOMMENDATIONS

Prepare a professional **3–5 page Business Insights Report** summarizing your findings.

Your report should include:

Executive Summary

Provide a concise overview of:

- The business problem.

- The objective of the analysis.
- The most important findings.

Key Analytical Findings

Summarize:

- Important patterns.
- Significant statistical findings.
- Key relationships.
- Important feature insights.

Business Implications

Explain what your findings mean for the organization or decision-makers.

Recommendations

Provide evidence-based recommendations based on your analysis.

Dataset Limitations

Discuss any limitations, such as:

- Data quality limitations.
- Small sample sizes.
- Missing information.
- Potential bias.
- Generalisation limitations.

Next Steps

Explain how the refined dataset can be used for predictive modelling in Week 4.

Deliverable

Submit a **Business Insights and Recommendations Report**.

5. MINIMUM PROJECT REQUIREMENTS

ADVANCED EXPLORATORY DATA ANALYSIS

Your Week 3 notebook should include:

- At least **12 meaningful visualisations**.
- Advanced distribution analysis.
- Bivariate analysis.

- Multivariate analysis.
- Correlation analysis.
- Group comparisons.
- Target-variable analysis, where applicable.

Do not simply repeat all Week 1 visualisations. Focus on deeper analysis and new insights.

STATISTICAL ANALYSIS

Your project must include:

- At least **four appropriate statistical analyses or tests**.
- Clearly stated hypotheses.
- Appropriate methodology.
- Test statistics where applicable.
- P-values where applicable.
- Clear interpretation.
- Business implications.

FEATURE ENGINEERING

Your project should include:

- At least **three meaningful engineered features**, where appropriate.
- A clear explanation of how each feature was created.
- Business justification.
- Potential analytical or modelling value.

If your dataset does not reasonably support three meaningful engineered features, explain the limitation and focus on the highest-quality features that can be justified.

FEATURE EVALUATION

Your analysis should include:

- Evaluation of original features.
- Evaluation of engineered features.
- Identification of redundant or highly correlated features.
- Feature retention or removal decisions.
- Clear justification for major decisions.

DOCUMENTATION

Your notebook should include:

- Clear Markdown explanations.
- Well-commented code.
- Professional formatting.
- Tables where appropriate.
- Clear interpretation of analytical and statistical findings.

- Business-focused explanations.

6. RECOMMENDED GITHUB REPOSITORY STRUCTURE

Continue using the same repository from Weeks 1 and 2.

Your project may now be organized as follows:

Your Week 3 work should build on your existing repository and clearly demonstrate the progression of your project from Week 1 through Week 3.

7. DELIVERABLES

By the end of Week 3, you should submit:

1. **Week 3 Jupyter Notebook (.ipynb)** or Google Colab Notebook link.
2. **Project Continuity Summary.**
3. **Statistical Analysis Summary.**
4. **Feature Engineering Documentation.**
5. **Feature Evaluation and Selection Summary.**
6. **Business Insights and Recommendations Report.**
7. **Updated Data Dictionary.**
8. **Final Modelling Dataset (.csv).**
9. **Updated README.md.**
10. **requirements.txt**, where applicable.
11. **Updated GitHub Repository Link.**
12. **Google Drive Folder Link**, where required.
13. **Completed Week 3 Submission Form**, provided by AnalystLab Africa.

Ensure that all files and links are accessible before submission.

Assessment Criteria

Criteria	Weight
Advanced Exploratory Data Analysis	20%
Statistical Analysis & Interpretation	25%
Feature Engineering	15%
Feature Evaluation & Selection	15%
Business Insights & Recommendations	15%
Documentation & Professionalism	10%
Total	100%

Professional Development Requirement

As part of building your professional Data Science portfolio, you should:

- Update your existing GitHub repository with your Week 3 work.
- Update your `README.md` to reflect the advanced analysis stage.
- Document your learning process and major analytical decisions.

- Share your project progress professionally on LinkedIn and X where appropriate.
- Highlight key analytical or statistical insights.
- Tag AnalystLab Africa where applicable.
- Use the hashtag **#AnalystLabAfrica**.

Your posts should focus on your learning, methodology, analytical thinking, and professional development.

Professional Branding Reminder

AnalystLab Africa encourages every intern to document their learning journey professionally on LinkedIn and X. Consistently sharing your work helps build your portfolio, demonstrates continuous learning, and increases your visibility to recruiters and potential employers.